Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Learner Registration Number

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Wednesday 7 January 2026

Afternoon (Time: 40 minutes)

Paper reference **31617H/1B**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided – *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
  – Section A: Structures and functions of cells and tissues (Biology)
  – Section B: Periodicity and properties of elements (Chemistry)
  – Section C: Waves in communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

P 8 4 0 1 5 R A 0 1 1 2

Pearson

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Figure 1 shows an electron micrograph of a Gram-negative bacterium.

A region of the bacterium is shown at a greater magnification.

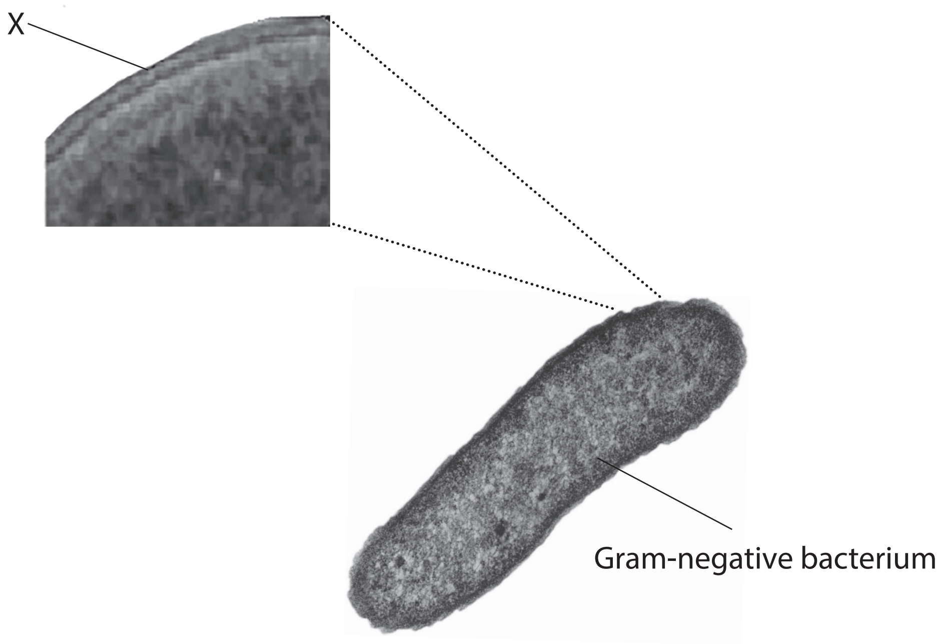

Figure 1

(a) Identify the structure labelled X in Figure 1.

(1)

☒ **A** cell wall

☒ **B** outer membrane

☒ **C** plasma membrane

☒ **D** plasmid

(b) State the colour a Gram-negative bacterium would be after Gram staining.

(1)

....

(c) Identify the cell structure present in a prokaryote cell.

(1)

- ☒ **A** centrioles
- ☒ **B** Golgi apparatus
- ☒ **C** nucleoid
- ☒ **D** tonoplast

An antibiotic can be used to treat an infection caused by Gram-positive bacteria.

(d) Explain the effect the antibiotic has on the Gram-positive bacteria.

(2)

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(Total for Question 1 = 5 marks)

2 A human brain contains 86 billion neurones.

The peripheral nervous system connected to the brain contains 15 billion neurones.

(a) Calculate the percentage increase in the number of neurones in the brain compared to the peripheral nervous system.

(3)

Show your working.

percentage increase = ...%

Figure 2 shows a synapse between two neurones.

Sodium channels are found on the post-synaptic membrane.

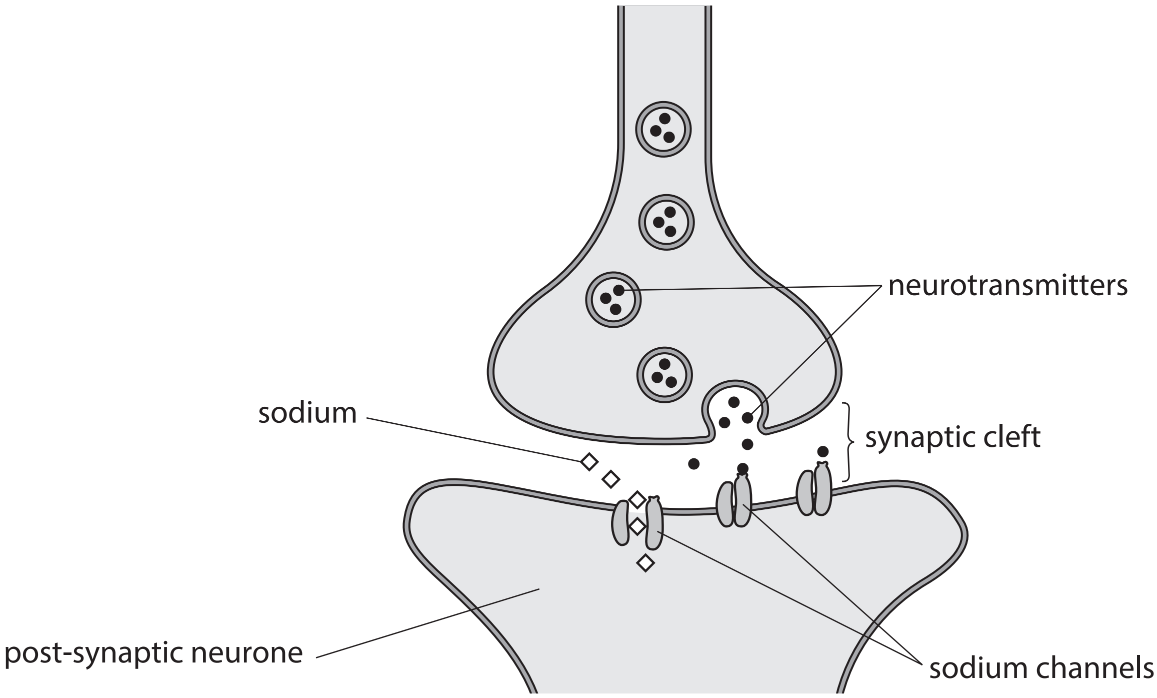

Figure 2

(b) Describe the role of the sodium channels in triggering an action potential.

(3)

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(Total for Question 2 = 6 marks)

3 Figure 3a shows a diagram of the lungs.

Figure 3b shows a healthy bronchus and the bronchus of a patient with chronic bronchitis.

Figure 3c shows healthy alveoli and the alveoli of a patient with emphysema.

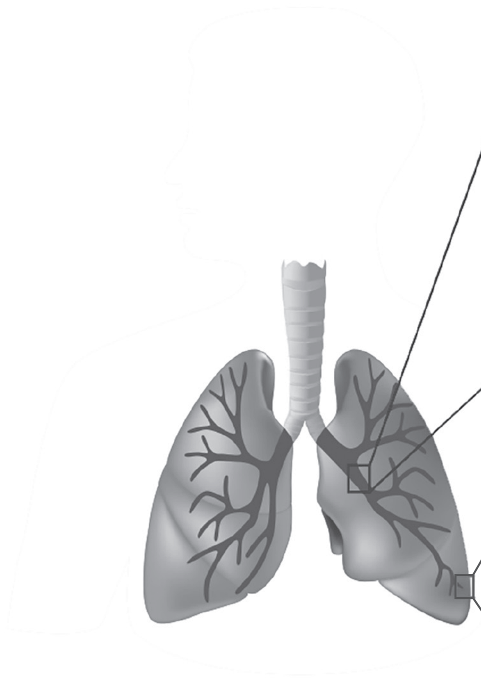

Figure 3a

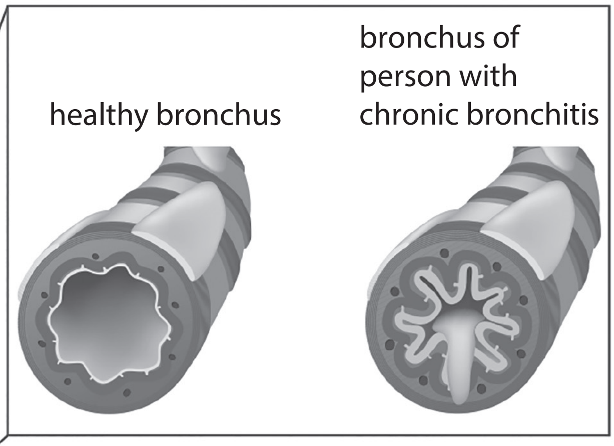

Figure 3b

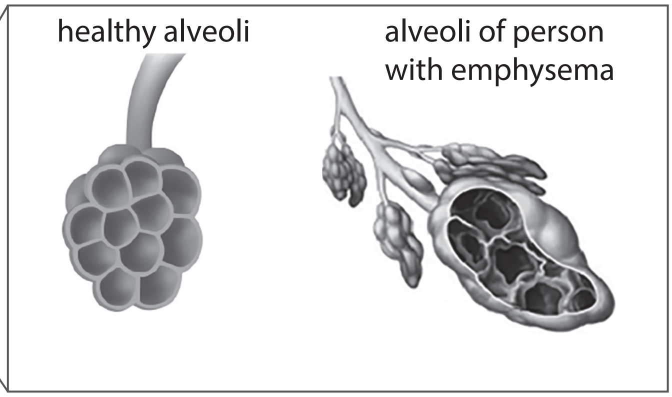

Figure 3c

(Source: ©PAL)

(a) Name the type of cells that line the alveoli.

(1)

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(b) State **two** symptoms of chronic bronchitis.

(2)

1 ....

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2 ....

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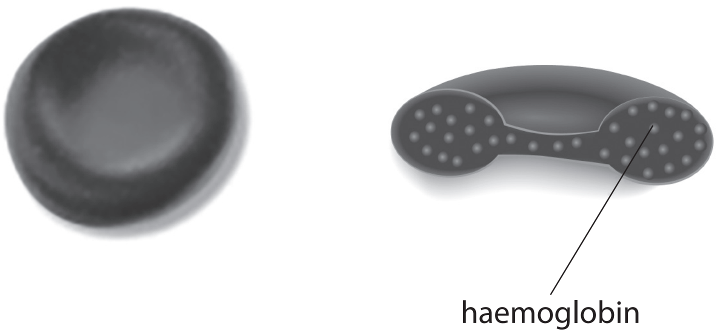
(c) Explain why a person with emphysema could find it difficult to exercise.

(4)

(Total for Question 3 = 7 marks)

4 Figure 4 shows a diagram of a red blood cell.



(Source: ©PAL)

Figure 4

Red blood cells are able to contain a lot of haemoglobin because they do not have a nucleus.

(a) Explain **one other** advantage of the lack of a nucleus in a red blood cell.

(2)

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Figure 5 shows the percentage saturation of haemoglobin with oxygen at different partial pressures of oxygen.

The saturation of haemoglobin increases as it binds oxygen.

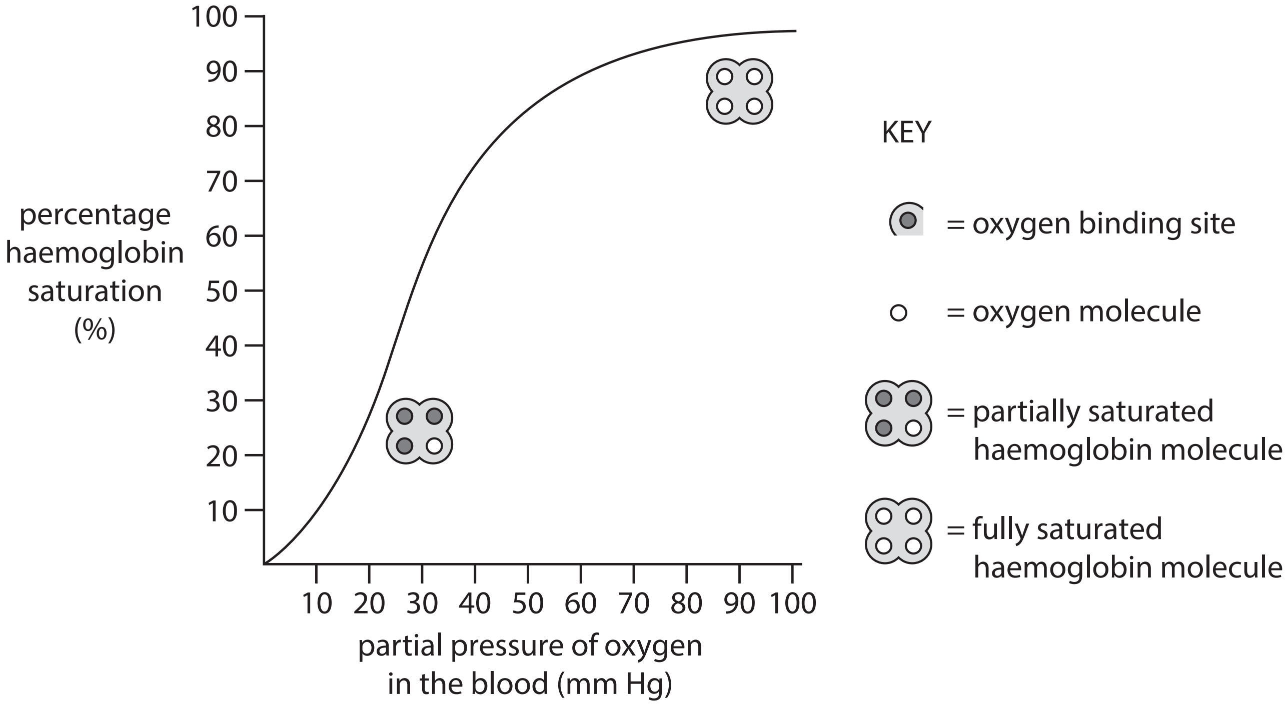

Figure 5

(b) Explain the effect of increasing partial pressure of oxygen on haemoglobin saturation.

Use information from Figure 5 to support your answer.

(4)

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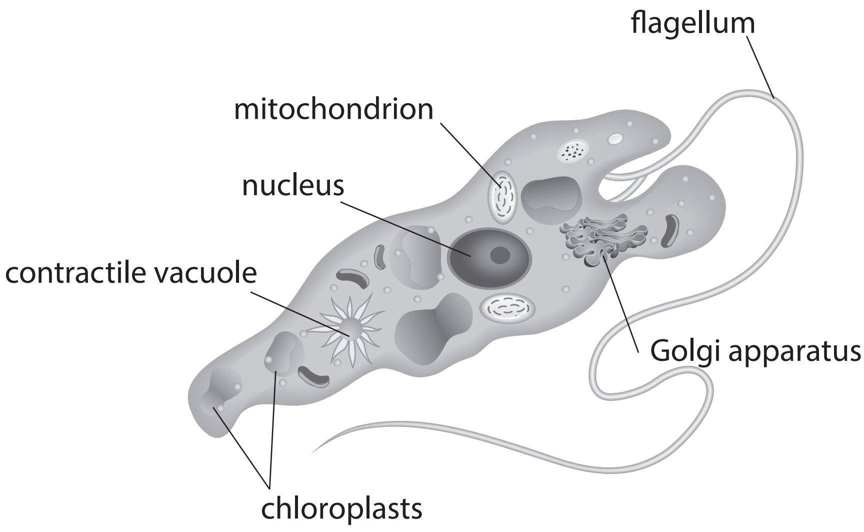
(Total for Question 4 = 6 marks)

5 Figure 6 shows a Euglena.

Euglena is a single cell organism that is difficult to classify.



(Source: ©PAL)

Figure 6

Discuss, using the information in Figure 6, why Euglena is difficult to classify.

(6)

You should include:

- what type of cell Euglena can be characterised as
- reference to both plant and animal cell characteristics.

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(Total for Question 5 = 6 marks)

TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS

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